

INTEGRATED TRANSCRIPTOMIC, DIFFERENTIAL EXPRESSION, PATHWAY ENRICHMENT, AND MACHINE LEARNING ANALYSIS OF BREAST CANCER MOLECULAR SUBTYPES

Using Orange Data Mining, GEO2R, Enrichr, STRING

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1. INTRODUCTION

Breast cancer is a very common and heterogeneous cancer among women globally. There are different types of breast cancer that have different biological characteristics, responses to treatment, and outcomes. One of the new bioinformatics tools used to gain insight into differences in the expression of genes across various cancer subtypes and to seek out possible biomarkers linked to disease state progression is gene expression analysis and understanding tumor biology.

The use of machine learning and transcriptomics has emerged as key computational tools in cancer research for discovering biomarkers, cancer classification, and precision oncology. Gene expression data constitute high dimensional molecular information which can be expressed by dimensionality reduction, clustering and supervised learning algorithms.

The project was focused on the gene expression dataset of breast cancer (**GSE45827**) obtained from the Gene Expression Omnibus (GEO) database/Kaggle and analyzed using Orange Data Mining and several external bioinformatics tools. This study aimed to explore transcriptomic variation among breast cancer molecular subtypes by preprocessing, Principal Component Analysis (PCA) and hierarchical clustering, feature selection, machine learning classification, differential expression analysis, pathway enrichment analysis, and protein interaction network analysis.

This was my first project involving transcriptomics and bioinformatics, but it was a very successful integration of machine learning and systems biology approaches to get biologically meaningful subtype associated biomarkers.

2. LITERATURE REVIEW

The molecular classification of breast cancer has greatly benefitted from the identification of gene expression-based signatures specific to the different subtypes. It has already been demonstrated that breast cancer can be classified into several molecular subtypes: HER2-positive, basal-like, Luminal A and Luminal B breast cancers, which have different biological and clinical properties.

For cancer subtype prediction and biomarker discovery, machine learning methods, like Support Vector Machine (SVM), Random Forest, and Logistic Regression have been extensively used in transcriptomic analyses. Other commonly used dimensionality reduction methods include Principal Component Analysis (PCA) and clustering algorithms, which can also be used to capture molecular heterogeneity and reveal biologically meaningful patterns of transcriptomics.

The impact of recently developed systems biology methods such as differential gene expression analysis, Gene Ontology (GO) enrichment, KEGG pathway analysis and STRING protein interaction networks has improved understanding of molecular mechanisms involved in tumor progression and their differences between tumor subtypes.

3. DATASET DESCRIPTION

In this study, we relied on the breast cancer transcriptomic dataset GSE45827, which was downloaded from the NCBI GEO Database repository.

The dataset contained:

- 151 biological samples
- Some 54,000 gene expression features

Samples were selected from a variety of molecular subtypes of breast cancer:

- HER2
- Basal
- Luminal A
- Luminal B
- Normal breast tissue
- Cell line samples

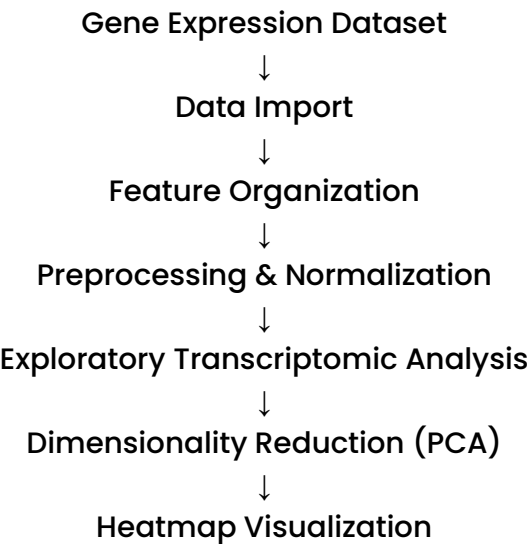
Data were obtained through transcriptomic profiling, with a microarray platform.

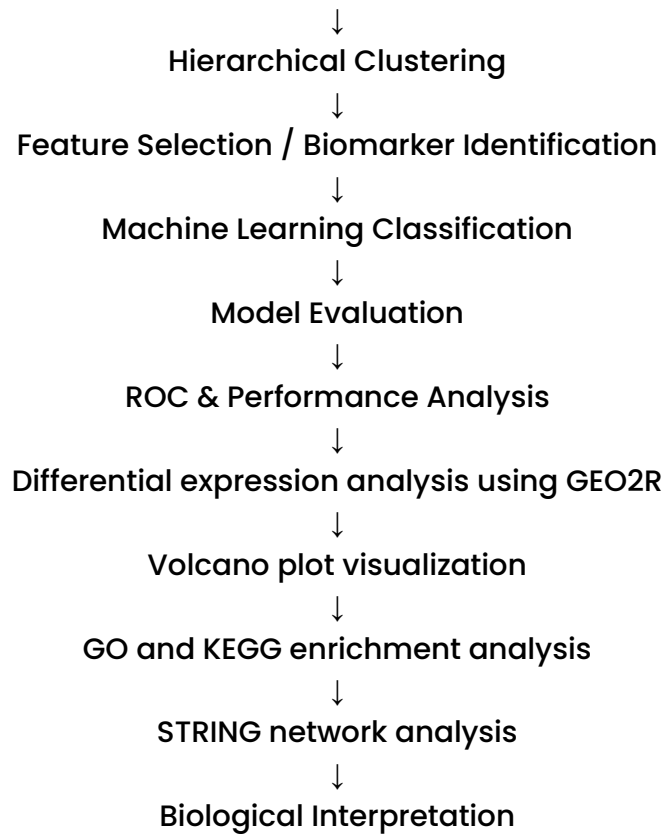
4. PLATFORMS AND BIOINFORMATICS TOOLS USED

Platform / Tool	Purpose
Orange Data Mining	Preprocessing, PCA, clustering, feature selection, machine learning, ROC analysis
NCBI GEO Database	Source of transcriptomic dataset (GSE45827)
GEO2R	Differential gene expression analysis
VolcanoR	Volcano plot visualization
Enrichr	GO Biological Process and KEGG pathway enrichment
STRING Database	Protein–protein interaction network analysis

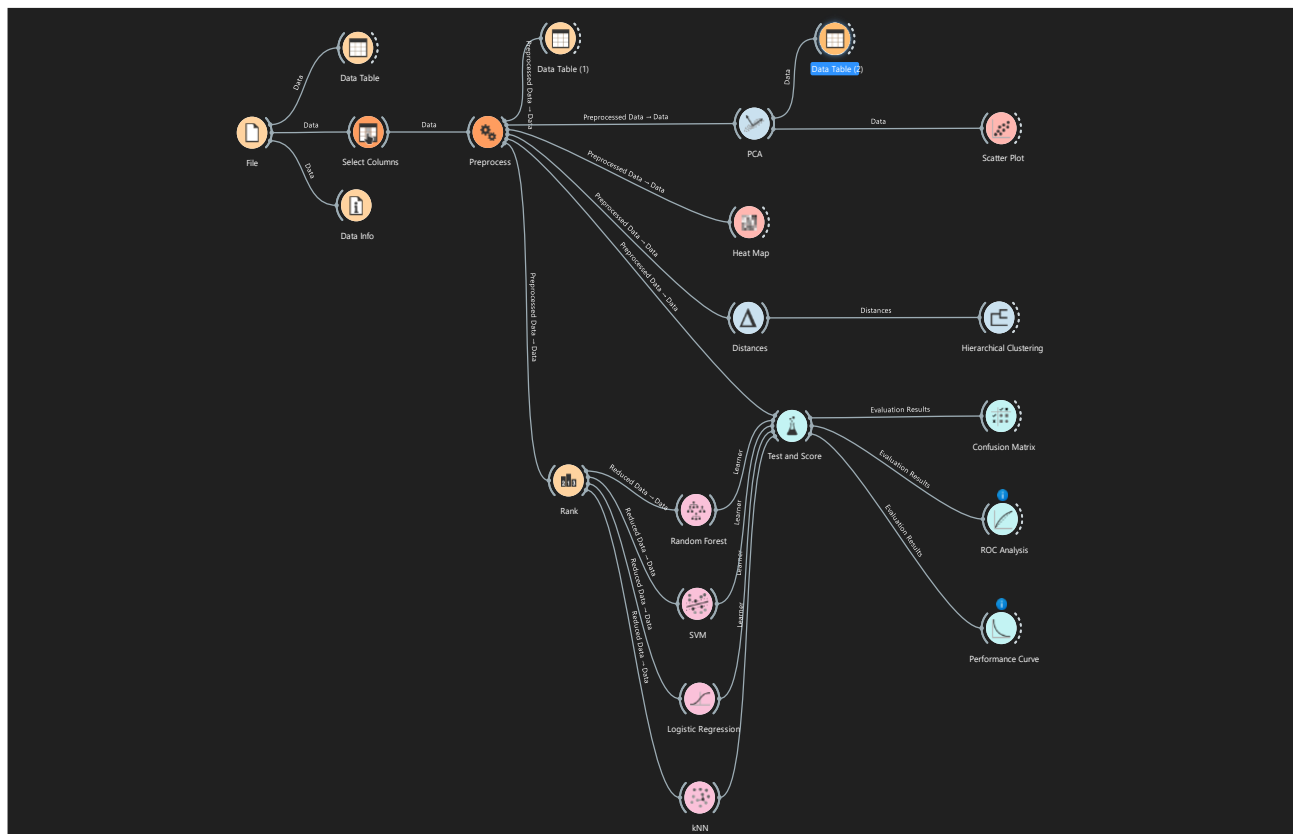
5. METHODOLOGY

Complete Gene Expression Analysis Workflow





This workflow combined transcriptomic analysis, machine learning and systems biology approaches.



The key elements of the Orange workflow were:

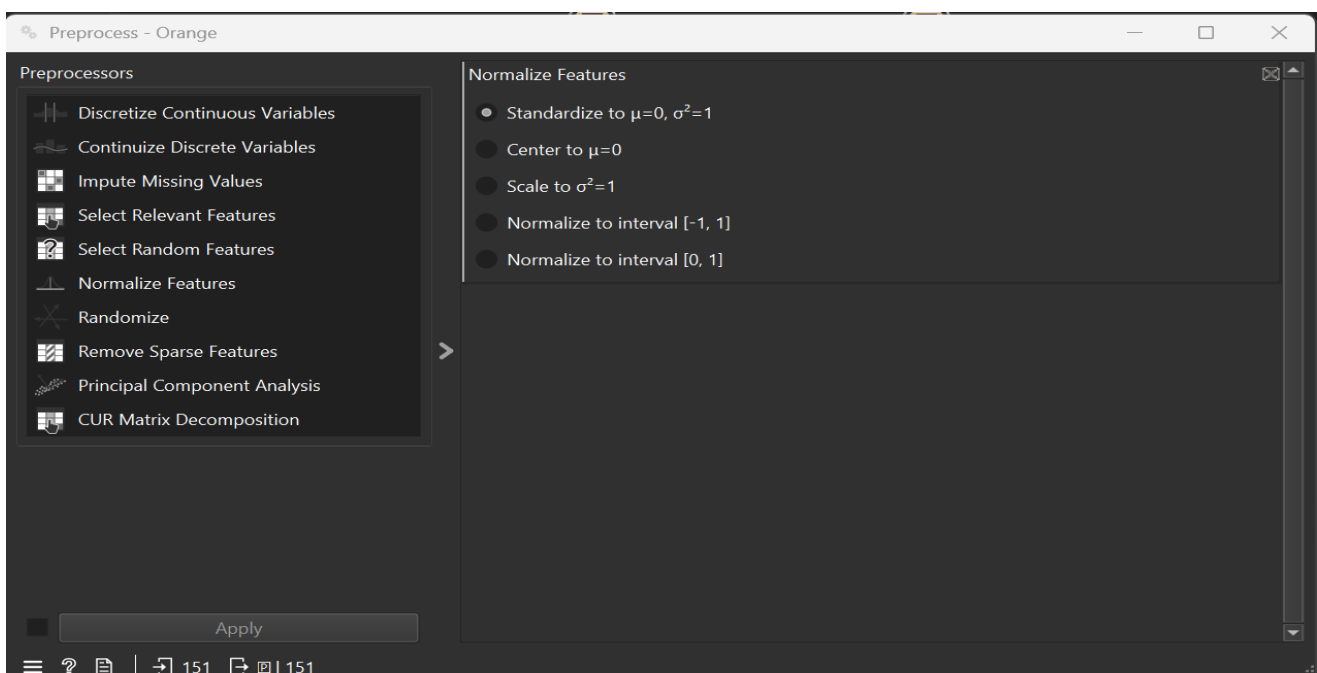
- File import
- Select Columns
- Preprocess
- PCA
- Heatmap
- Distances
- Hierarchical Clustering
- Rank
- Machine Learning Models
- Test & Score
- ROC Analysis
- Performance Curve

6. PREPROCESSING

A gene expression preprocess was done using Preprocess widget in Orange Data Mining. Normalization was performed, to minimize variations in gene expression values and to flatten the distribution of transcriptomes among samples.

To remove technical variability and differences in gene scale from the raw microarray data, normalization is an important step in gene expression analysis. The preprocessing was beneficial for the downstream analyses and for machine learning performance.

The normalized heatmap revealed more consistent expression patterns, which were also biologically relevant and captured the inter-subtype variability in the expression data.



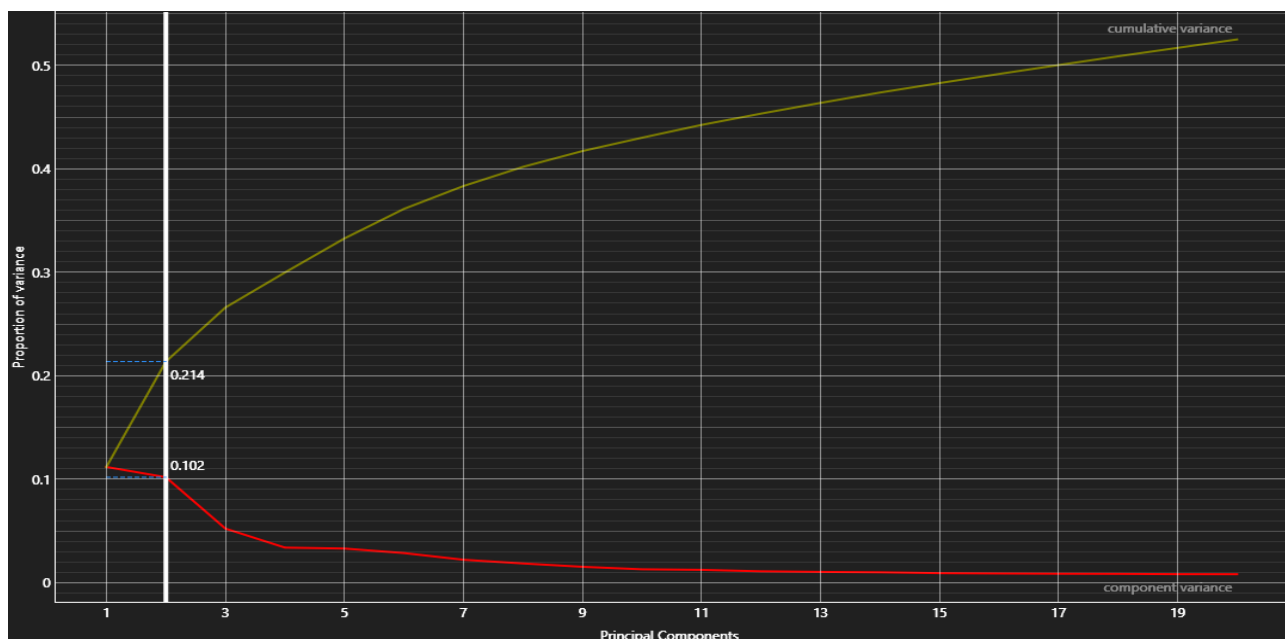
7. PCA ANALYSIS

To visualize the molecular variation between the different types and reduce dimensions of the high throughput transcriptomic data we performed Principal Component Analysis (PCA).

The two first principal components accounted for about 21% of the total variance of the dataset. Clustering patterns of subtype-specific samples were observed in a PCA scatter plot visualization of HER2, basal, luminal_A, luminal_B, normal and cell line samples.

Normal samples from breast tissue clustered into a very distinct group that was readily distinguishable from the breast tumor groups, suggesting significant molecular differences between normal and cancerous tissues. Samples from cell lines were also highly separated, which was probably because of transcriptomic variations in conditions of in vitro culture.

A greater transcriptomic dispersion was observed in aggressive tumour types, basal-like cancer and HER2 positive, which correlated with an increase in molecular heterogeneity.



Analysis of Gene Expression Profiles in Breast Cancers Is Subtype-Specific :

1. HER2 Subtype Analysis

Samples positive for HER2 were grouped in a moderate cluster that was mainly located in the central area of the PCA plot. Some overlap in samples was observed with the luminal subtypes and hence, there were some common transcriptomic features involved in the proliferative signaling pathway. Notably, there was also significant scatter seen in the HER2 samples, suggesting that there is molecular heterogeneity within the HER2 subtype. Proliferation-associated markers like PLK1, TPX2 and CDC20 are highly expressed, indicating high mitotic activity and aggressive tumor biology. The PCA

distribution is designed to reflect the biology of HER2-positive breast cancers, which are known to be highly proliferative tumors with dependence on HER2 mediated oncogenic signaling.

2. Basal Subtype Analysis

Basal-like breast cancer samples clustered fairly broadly, and mostly in the lower PC2 region of the PCA space. The higher proportion of the basal subtype suggests there is a significant transcriptomic diversity in this subtype. Basal tumors are characterized by a high proliferation rate, genomic instability, and aggressive clinical behavior, all of which are supported by the high expression of genes involved in the mitotic checkpoint process, including BUB1, ASPM, TTK and CDC20. Separation of basal samples from luminal and normal breast groups reveals that there are significant differences in the molecules between hormone receptor-negative and hormone receptor-positive breast cancers.

3. Luminal A Subtype Analysis

Luminal A samples clustered rather compactly and organically in the upper-central portion of the PCA region. The high degree of clustering suggests higher transcriptomic similarity than aggressive tumor types. Luminal A tumors are generally more likely to be hormone receptor positive, have less proliferation and a favorable clinical outcome. These biologically stable molecular properties are reflected in the clustering of the cases, as seen in the PCA plot. Lesser spread of Luminal A samples indicates lower gene expression variability and more homogenous gene expression between samples.

4. Luminal B Subtype Analysis

Luminal B samples were located near Luminal A clusters, but had slightly larger dispersion patterns. This suggests a degree of transcriptomic similarity between the two luminal subtypes, but also suggests a greater proliferation in Luminal B tumors. Luminal B tumours are typically more aggressive than Luminal A tumours and have an increased expression of cell cycle genes. The transitional molecular features of this subtype are confirmed by the intermediate clustering seen in the PCA plot.

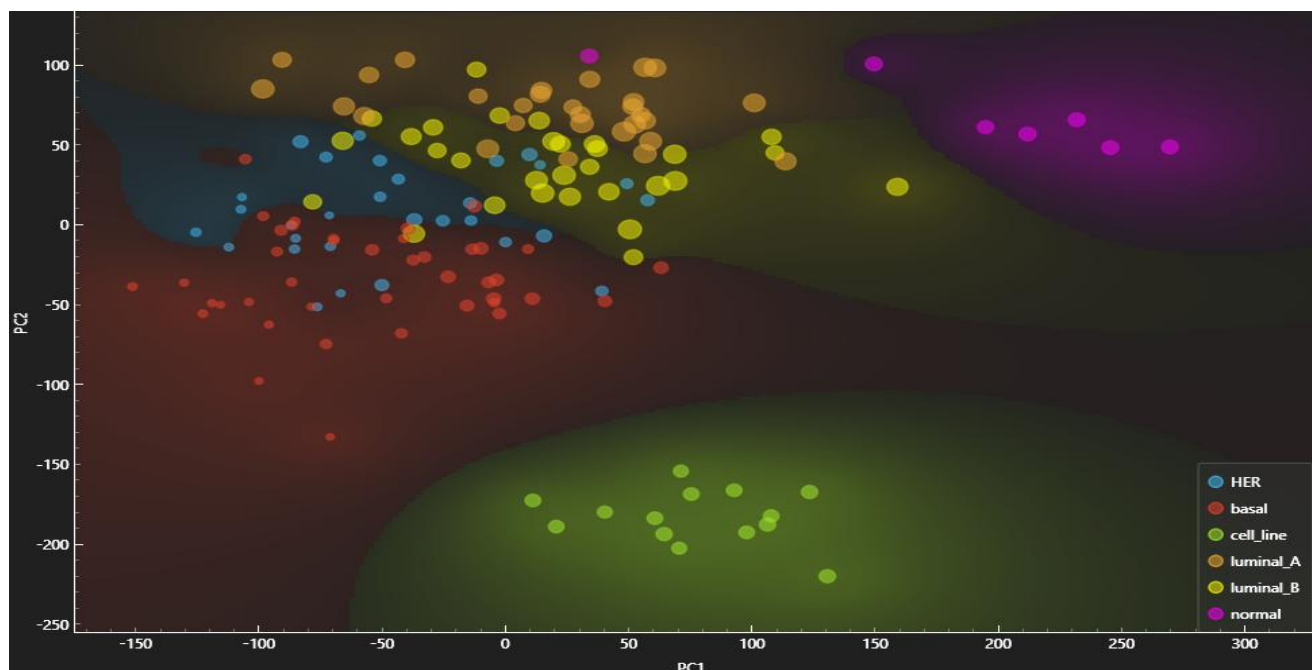
5. Normal Breast Tissue Analysis

Numerous samples from normal breast tissues were located in a distinct and well-segregated area in the right-most area of the PCA plot. Many samples from normal breast tissue were clustered into a well-segregated area in the right-most quadrant of the PCA plot. This separation is significant and shows that there is a large difference between the transcriptomic characteristics of healthy breast tissue and the tissue of tumors. High molecular consistency of normal samples is shown by the compact clustering pattern. The clear separation from cancer subtypes was possibly partly due to the low expression

of proliferation-associated biomarkers and to the lack of aggressive oncogenic signatures.

6. Cell Line Sample Analysis

Samples of the cell lines were highly isolated in the lower right PCA region, indicating high transcriptomic distance from patient derived tumour tissues. This separation is probably due to changes at the molecular level resulting from adaptation to long-term in vitro culture and artificial growth conditions. The proliferation, metabolic and stress-response pathways of cell line samples can differ from clinical tumor samples. The distinct clustering behavior agree with the fact that in vitro models have a transcriptomic profile that differs significantly from primary breast cancer tissue.



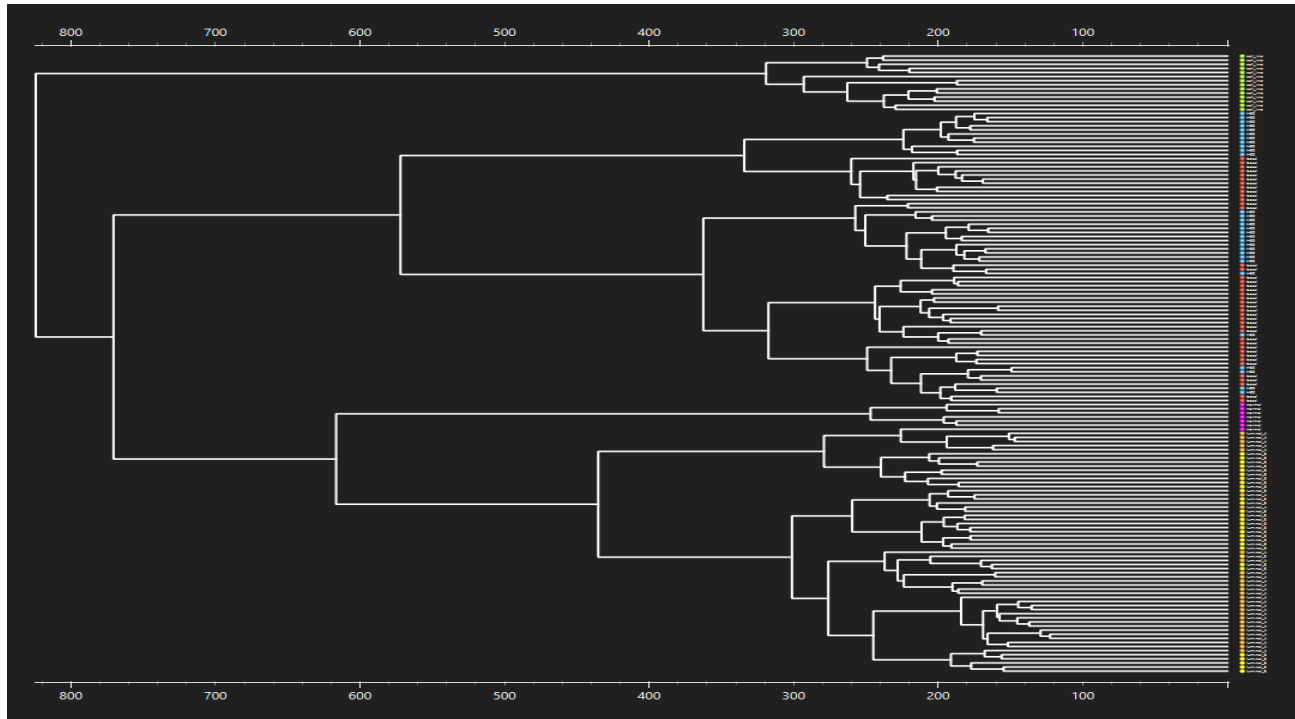
8. CLUSTERING ANALYSIS

Normalized Euclidean distance and Ward linkage were used for hierarchical clustering analysis. The dendrogram showed that there were grouping patterns among breast cancer molecular subtypes that were biologically meaningful.

The samples in Luminal A and Luminal B were grouped together into relatively compact clusters, indicative of the hormone receptor-associated transcriptomic signatures shared between these groups. Basal-like tumors exhibited greater dispersion with more molecular heterogeneity and aggressive biological behavior.

The breast normal tissue samples were grouped together separately from the tumor associated groups, indicating significant transcriptomic differences between normal and tumor tissue.

The clusters generated by the ward linkage method were very compact and separated from each other, providing it was very well suited for transcriptomic subtype analysis.



9. FEATURE SELECTION

To develop subtype-associated transcriptomic biomarker, the Rank widget was used to perform feature selection. Various scoring methods were employed to rank genes based on their significance for classification of the sub-types, such as Information Gain, Gain Ratio, Gini Decrease and ANOVA.

The following genes were among the top ranked genes:

- PLK1 - 202240_at
- TPX2 - 210052_s_at
- CDC20 - 202870_s_at
- BUB1 - 209642_at
- ASPM - 219918_s_at
- KIF14 - 236641_at
- TTK - 204822_at
- JMJD2B - 1558448_a_at

Many of these genes are associated with:

- mitotic regulation
- cell cycle progression
- chromosomal instability

- tumor proliferation

These biomarkers had good discriminatory ability between breast cancer molecular subtypes.

Rank - Orange

Scoring Methods		#	Inf...ain	Gain ratio	Gini	ANOVA
☒ Information Gain	1	N 202240_at	1.048	0.524	0.258	94.431
☒ Information Gain Ratio	2	N 211519_s_at	1.038	0.519	0.246	134.626
☒ Gini Decrease	3	N 215397_x_at	0.998	0.499	0.219	73.148
☒ ANOVA	4	N 210052_s_at	0.993	0.497	0.228	153.135
☐ χ^2	5	N 1558448_a_at	0.969	0.485	0.218	98.836
☐ ReliefF	6	N 209642_at	0.964	0.482	0.226	121.081
☐ FCBF	7	N 219918_s_at	0.964	0.482	0.238	131.557
	8	N 236641_at	0.963	0.481	0.231	103.128
	9	N 204822_at	0.960	0.480	0.243	112.183
	10	N 202870_s_at	0.947	0.474	0.219	108.148
	11	N 209408_at	0.944	0.472	0.215	132.053
	12	N AFFX-r2-P1-...	0.941	0.470	0.186	110.763
	13	N 204162_at	0.939	0.470	0.210	105.634
	14	N 202705_at	0.932	0.466	0.221	115.954
	15	N 213226_at	0.930	0.465	0.222	116.092
	16	N 203554_x_at	0.929	0.465	0.221	97.262
	17	N 209464_at	0.925	0.463	0.224	119.332
	18	N 200934_at	0.925	0.462	0.216	85.492
	19	N 204962_s_at	0.925	0.462	0.229	119.919
	20	N 220651_s_at	0.923	0.462	0.224	100.652

Select Attributes

☐ None

☐ All

☐ Manual

☒ Best ranked: 100

☒ Send Automatically

151 | 151 | 54.7k | 100

10. MACHINE LEARNING CLASSIFICATION

A classification of breast cancer molecular subtypes was performed using multiple supervised machine learning algorithms using transcriptomic biomarkers.

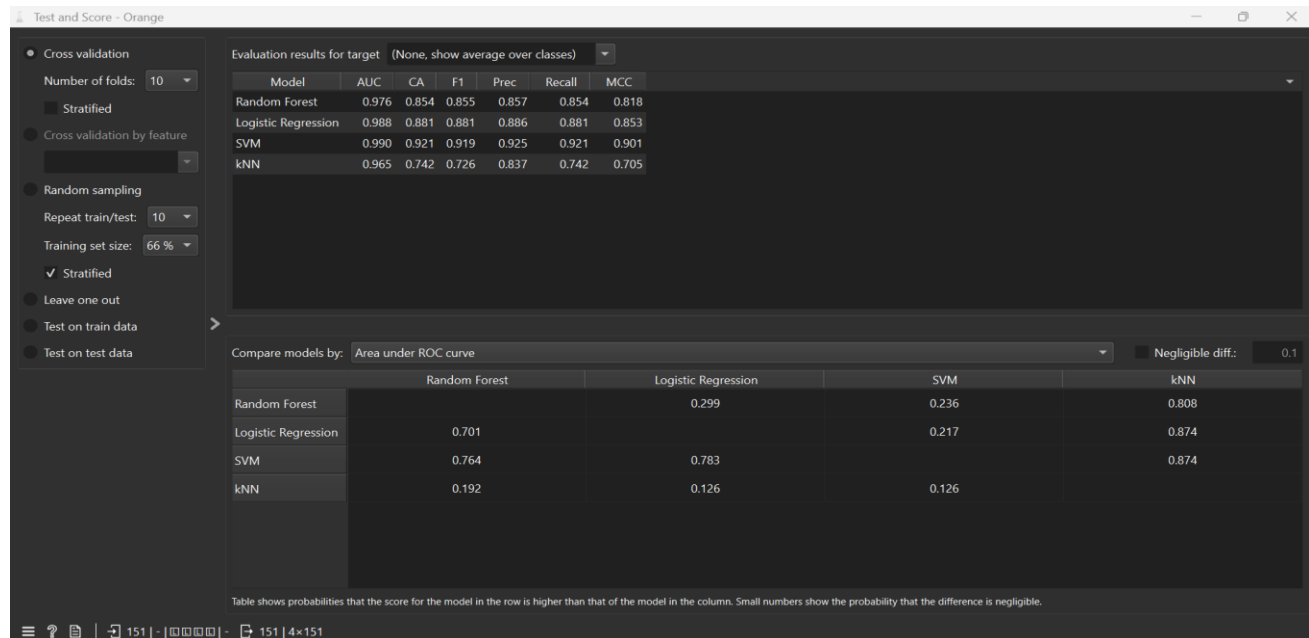
Model	AUC	CA	F1 Score	Precision	Recall	MCC	Performance Interpretation
Random Forest	0.976	0.854	0.855	0.857	0.854	0.818	Strong ensemble classifier with robust performance on high-dimensional gene expression data
Logistic Regression	0.988	0.881	0.881	0.886	0.881	0.853	Demonstrated highly stable and balanced subtype classification performance
SVM	0.990	0.921	0.919	0.925	0.921	0.901	Best-performing classifier with excellent discrimination of breast cancer subtypes
kNN	0.965	0.742	0.726	0.837	0.742	0.705	Lower performance likely due to sensitivity to high-dimensional transcriptomic features

The following classifiers were evaluated:

- Random Forest
- Support Vector Machine (SVM).

- Logistic Regression
- k-Nearest Neighbor (kNN)

Top 100 genes were selected based on features, and subsequently fed into the classification analysis.



SVM showed the highest classification accuracy for all models as:

- AUC = 0.990
- Accuracy = 92.1%
- F1-score = 0.919

Logistic Regression and Random Forest yielded excellent prediction performance, and kNN had relatively poor classification accuracy.

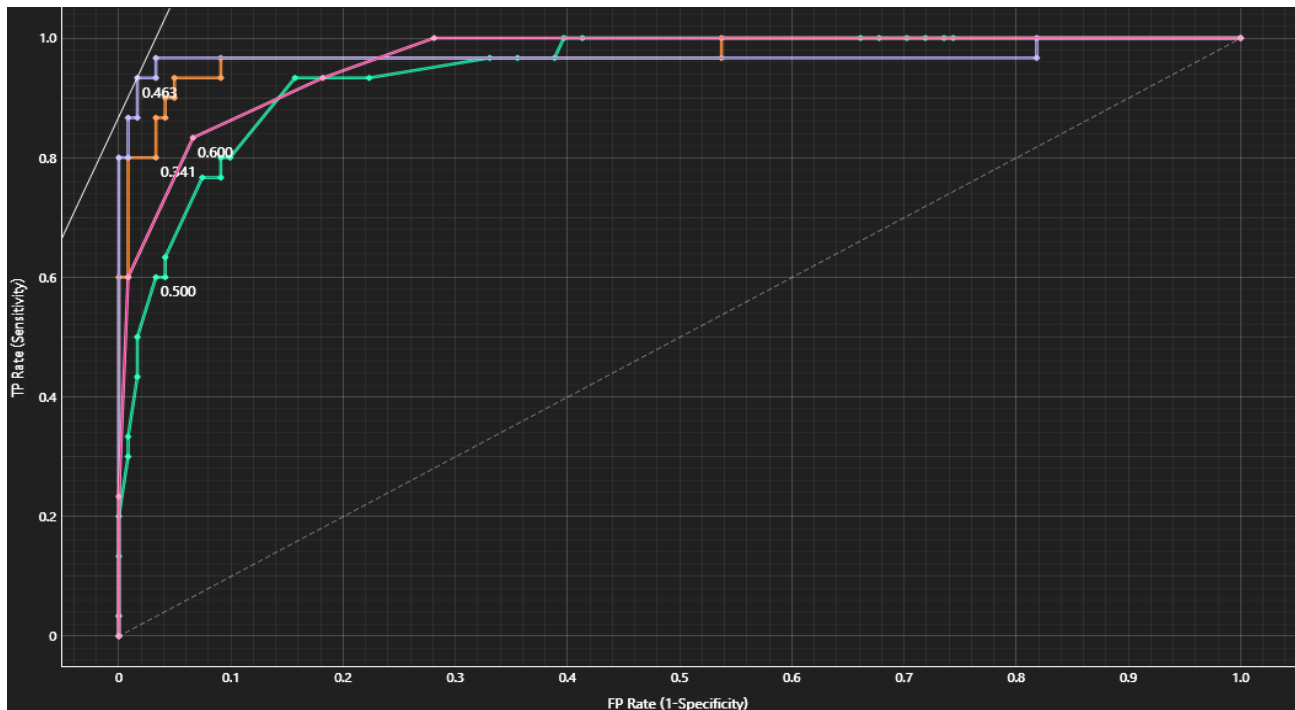
From the results, it can be concluded that the transcriptomic signatures of the different breast cancer molecular subtypes are very distinct.

II. ROC EVALUATION

The machine learning classifiers were assessed using ROC analysis. The ROC curves of the SVM, LR and RF models were also well clustered near the upper-left corner of the plot, suggesting high sensitivity and specificity in the classification of subtypes.

SVM performed best with the highest AUC value indicating that it had the best classification ability among all the classifiers to classify breast cancer subtypes by using transcriptomic biomarkers.

The predictive efficiency was also excellent, as demonstrated by performance curve analysis at various probability thresholds.



12. DIFFERENTIAL GENE EXPRESSION ANALYSIS

To compare the gene expression levels of breast cancer samples that are HER2 positive with normal breast tissue samples, the differential gene expression analysis was done externally with GEO2R. VolcanoNoseR was used to create a volcano plot to represent significantly altered genes. The volcano plot showed a significant difference in transcription between HER2-positive tumors and normal breast tissue. Many genes were found to be significantly under-expressed and over-expressed, suggesting extensive molecular disturbances linked to progression of the tumour.

Genes with high significance were :

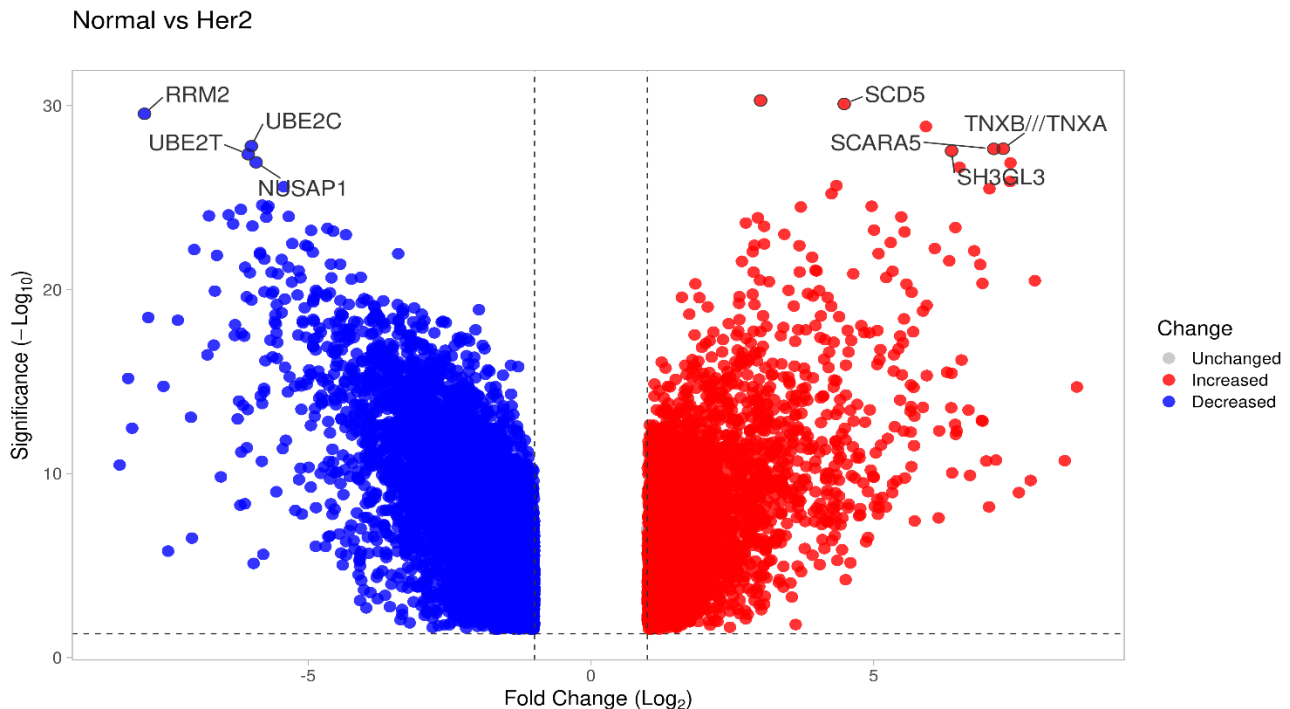
Upregulated genes :

- SCD5
- SCARA5
- TNXB/TNXA
- SH3GL3

Downregulated genes :

- RRM2
- UBE2C
- UBE2T
- NUSAP1

The good separation of differentially expressed and unchanged genes was indicative of good biological separation between tumor and normal samples.



13. GO BIOLOGICAL PROCESS ENRICHMENT ANALYSIS (Enrichr)

Enrichr was used for GO Biological Process enrichment analysis.

The biomarker genes identified in the enrichment analysis were highly enriched for the following pathways:

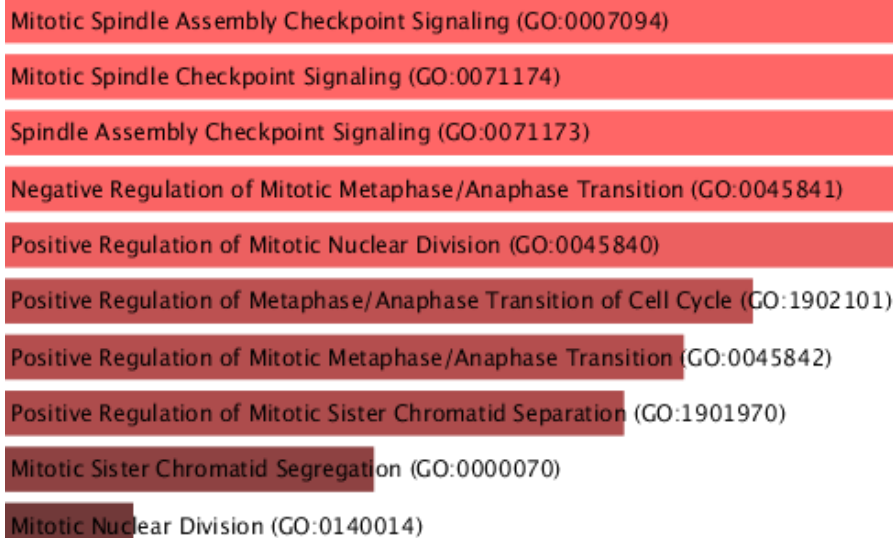
- mitotic spindle checkpoint signaling
- chromosome segregation
- metaphase/anaphase transition
- mitotic nuclear division
- sister chromatid segregation
- The signaling at the spindle assembly checkpoint.

They had highly enriched genes for:

- PLK1 - 202240_at
- CDC20 - 202870_s_at
- BUB1 - 209642_at
- TPX2 - 210052_s_at
- TTK - 204822_at
- UBE2C - 202954_at

These pathways are crucial for the correct functioning of mitosis and genomic integrity.

The results show a strong association between dysregulated machinery of the mitotic checkpoints and unchecked proliferative signaling in HER2-positive breast cancer.



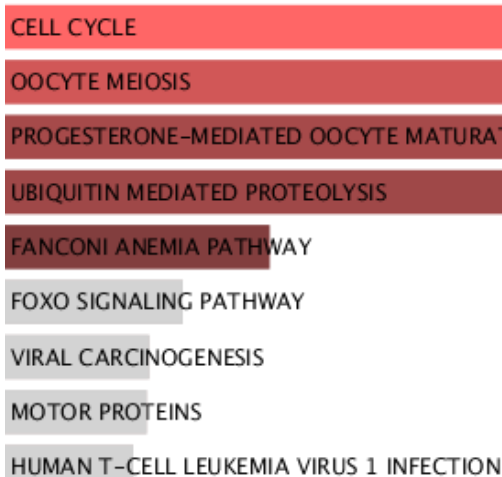
14. KEGG PATHWAY ANALYSIS

KEGG pathway enrichment analysis showed that the pathways related to tumour proliferation and cell cycle regulation were significantly enriched.

The most enriched pathways were:

- Cell cycle
- Oocyte meiosis
- Progesterone-mediated oocyte maturation
- Ubiquitin-mediated proteolysis
- FOXO signaling pathway

The cell cycle-related pathways involving PLK1, CDC20, TTK and BUB1 are enriched and suggest greater proliferation activity in HER2-positive tumors. The results further support the involvement of mitotic dysregulation and checkpoint abnormalities in breast cancer progression.



15. STRING PROTEIN INTERACTION NETWORK ANALYSIS

The identified biomarkers showed high functional interaction between one another in STRING protein–protein interaction analysis.

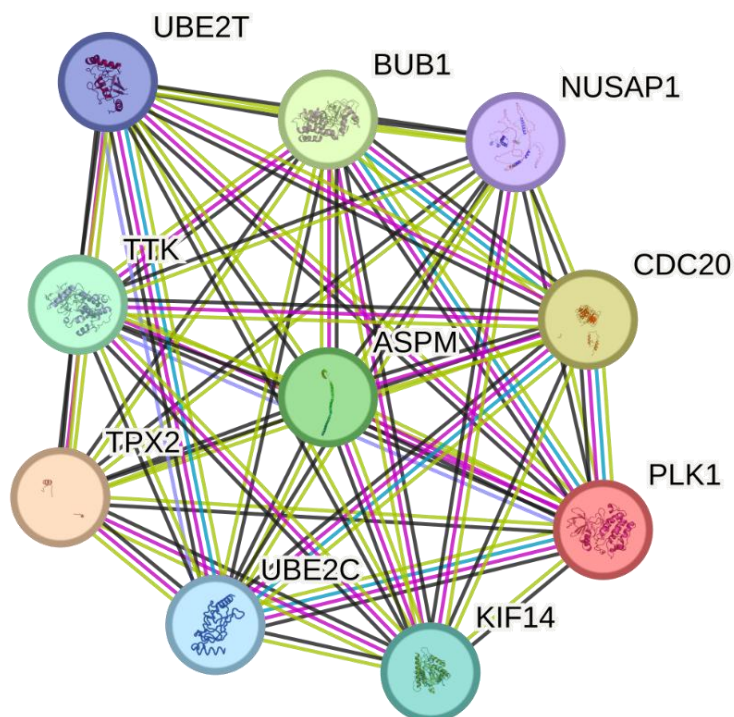
Key hub genes were found to be:

- PLK1 - 202240_at
- CDC20 - 202870_s_at
- TPX2 - 210052_s_at
- BUB1 - 209642_at
- ASPM - 219918_s_at
- TTK - 204822_at
- UBE2C - 202954_at
- KIF14 - 236641_at

A very connected network of genes involved in proliferation and mitogenic signaling was created.

The high interaction density suggests that the identified biomarkers are interconnected, operating as part of a coordinated system of mitotic checkpoint and chromosome segregation machinery, and not as independent genes.

The selected transcriptomic biomarkers were also shown to have biological relevance by the network analysis.



16. BIOLOGICAL INTERPRETATION

The integrated transcriptomic analysis confirmed that there are distinct molecular differences between breast cancer subtypes. Basal-like and HER2 positive tumor groups showed high expression of cell cycle genes like PLK1, TPX2, CDC20, and BUB1.

Luminal subtypes were relatively monomodular and had lower proliferative properties. Strong molecular separation was observed between the normal breast groups and the tumor associated groups in normal breast tissue samples.

The biomarkers identified are associated with:

- mitotic progression
- spindle checkpoint regulation
- chromosomal instability
- cell cycle regulation
- tumor proliferation

These genes were further confirmed to play a critical role in tumor progression during HER2-associated tumor pathogenesis through GO enrichment, KEGG pathway analysis and STRING interaction analysis.

The results indicate that abnormal activation of signaling pathways controlling proliferation and mitogenesis plays an important role in the aggressive behaviour of breast cancer cells.

Top 10 Genes from Rank Widget

1. PLK1 (Polo-Like Kinase 1) - 202240_at

PLK1 is a serine/threonine kinase which is essential for cell cycle regulation in the processes of mitosis and chromosome segregation. PLK1 is often up-regulated in breast cancer and has been found to be tightly linked to high tumour proliferation and poor prognosis. High levels of PLK1 have been seen in aggressive types of breast cancer like basal-like and HER2 positive because of their high mitotic activity. From the clinical point of view, PLK1 is deemed a significant therapeutic target since PLK1 inhibition may inhibit the growth of tumor cells and lead to apoptosis. The cell cycle regulation pathway comes up high in this analysis, indicating the importance of cell cycle deregulation as a significant molecular characteristic of breast cancer subtypes.

2. KNSL6 (Kinesin-Like Protein 6) - 211519_s_at

KNSL6 is a member of the kinesin family of motor proteins that are used for intracellular transport and cell division spindle organization. A few cancers, including breast cancer, have been known to have abnormal expression of genes related to kinesin. Enhanced activity of KNSL6 could be linked to further success in the process of mitosis and genomic instability in highly proliferative tumours. It is expressed especially in more aggressive breast cancer types where there is a quick growth in the cells. In the clinic,

kinesin family proteins are being explored as therapeutic targets for the development of anti-cancer drugs since disruption of transport systems during mitosis will have an effect on the progression of tumors.

3. THSD4 (Thrombospondin Type 1 Domain Containing 4) - 215397_x_at

The extracellular matrix organization and cell adhesion processes in which THSD4 is involved play a role in tissue remodeling and interactions with the tumor microenvironment. In breast cancer, the changes in extracellular matrix signaling contributes to tumour invasion, migration and metastasis potential. Increased and decreased expression of THSD4 could be indicative of alterations in stromal interactions of different breast cancer subtypes. Extracellular matrix remodeling signatures are common in both basal-like and invasive tumors, and these can drive metastasis and/or tumor aggressiveness. From a clinical perspective, THSD4 related pathways could be important in the understanding of tumor progression and could be used as markers of the invasive behaviour of the disease.

4. TPX2 (Targeting Protein for Xklp2) - 210052_s_at

TPX2 is a protein that is part of a group of proteins called microtubular associated proteins, which play a role in spindle assembly and progression through mitoses during cell division. TPX2 is reported to be over-expressed in breast cancer and has been shown to be strongly linked to enhanced proliferation, chromosomal instability and more aggressive behaviors of tumor cells. Breast cancer cells with high TPX2 expression are found in the basal and HER2-positive breast cancer subtypes, which are associated with increased cell division. In clinical terms, TPX2 is clinically significant as it is frequently associated with worse prognosis and increased metastatic potential. With such a high ranking feature, the dysregulation of mitosis is likely to be an important discriminating feature between breast cancer molecular subtypes.

5. JMJD2B (Jumonji Domain Containing 2B) - 1558448_a_at

JMJD2B is a histone demethylase that plays a role in epigenetic regulation and chromatin remodeling. In breast cancer, JMJD2B is associated with the transcriptional activation of pathways that are involved in the pathogenesis of the disease, including oncogenic pathways and hormones responsive genes. It is specifically linked to the estrogen receptor-positive luminal breast cancer cell types and is known to support growth and survival in these cell types via epigenetic modulation. The upregulation of JMJD2B has been associated with greater proliferation and survival in the presence of hypoxia within tumour cells. From a clinical point of view, JMJD2B is a promising epigenetic therapeutic target because it is involved in the regulation of transcription of cancer associated pathways.

6. BUB1 (Budding Uninhibited by Benzimidazoles 1) - 209642_at

BUB1 is a mitotic checkpoint kinase that plays roles in the alignment of chromosomes and regulation of the spindle checkpoint in cell division. BUB1 has been shown to be overexpressed in breast cancer and is linked to chromosomal instability, uncontrolled proliferation and unfavorable prognosis in breast cancer patients. Breast cancer subtypes that are more aggressive tend to have more BUB1 expression due to more genomic instability and more mitotic activity. BUB1 is clinically recognised as a possible prognostic factor and therapeutic target in cancers with proliferation. This paper's powerful contribution in this analysis underlines the significance of mitotic checkpoint abnormality in the classification of breast cancer subtypes.

7. Abnormal Spindle Microtubule Assembly (ASPM) - 219918_s_at

During cell division, ASPM is involved in the organization of the cell's mitotic spindle and correct segregation of the chromosomes. ASPM has been linked to aggressive tumour growth and increased proliferation in breast cancer. Basal-like and HER2+ tumors often have increased ASPM, being highly proliferative. Overexpression of ASPM has also been associated with poor prognosis in patients and increased metastatic properties. In a clinical setting, ASPM is becoming a candidate for tumor aggressiveness and as a target for anti-proliferative therapy.

8. KIF14 (Kinesin Family Member 14) - 236641_at

KIF14 is a kinesin motor protein which functions in the regulation of cytokinesis, chromosome segregation and the progression of the cell cycle. KIF14 is often amplified and over-expressed in breast cancer and linked to a poor prognosis and aggressive tumour phenotypes. Rapidly proliferating subtypes like basal and HER2-positive express high levels of KIF14. In the clinic, KIF14 has attracted attention as a prognostic biomarker since it is clearly associated with tumor progression and metastasis. This importance in this analysis implies that there are some significant differences in the transcriptomic profiles between subtypes which are due to the action of the mechanisms of control of mitosis.

9. TTK (TTK Protein Kinase) - 204822_at

TTK is a dual specificity protein kinase that is associated with signaling and maintenance of the spindle assembly checkpoint (SAC) and chromosomal stability in the course of cell division. TTK is often upregulated in highly proliferating and genomically instability tumors in breast cancer. The basal-like and triple-negative breast cancers typically have high levels of TTK which are associated with fast mitogenic activity. Cytologically, TTK inhibitors are currently being investigated as anti-cancer drugs, since TTK inhibition can cause mitotic catastrophe and cell death in cancer cells. This study

showed that it ranks strongly in this study, suggesting its significant role in the proliferation of subtypes associated signaling.

10. CDC20 (Cell Division Cycle 20) - 202870_s_at

CDC20 is an important regulatory protein for the promotion of anaphase through the activation of anaphase-promoting complex. Overexpression of CDC20 is clearly linked with high tumor proliferation and chromosomal instability as well as poor prognosis in breast cancer. A high level of expression of CDC20 is often seen in highly aggressive breast cancer subtypes, because they often have increased mitotic demands. From a clinical standpoint, CDC20 is a candidate for being a useful biomarker of tumour aggressiveness and as a therapeutic target for blocking unchecked cell division. This project assigned it a high importance score due to the fact that cell cycle dysregulation plays a crucial role in molecular subtyping of breast cancer.

17. DISCUSSION

In this project, the successful integration of transcriptomics, machine learning and systems biology approaches for breast cancer subtype characterization was successfully demonstrated.

There was evidence of subtype-specific molecular signatures by PCA and clustering. Highly informative biomarkers were identified by feature selection and machine learning classification that are able to accurately distinguish breast cancer subtypes.

The addition of differential expression analysis, GO enrichment, KEGG pathways, and STRING interaction analysis greatly enhanced the biological interpretation of this study.

Main points descriptive of the biological reliability of the workflow were that the identified biomarkers were highly correlated with the mitotic checkpoint signaling and the proliferative regulation.

This was the first time that I had worked through a transcriptomics and bioinformatics project; however, the integration of several cutting-edge analytical methods into a unified cancer systems biology study was successful.

18. LIMITATIONS

During the study, the following limitations were noted :

- relatively small sample size
- high-dimensional transcriptomic complexity
- Lack of external validation sets
- limited differential expression analysis within Orange Bioinformatics
- Lack of survival and clinical outcome analysis

Furthermore, a high number of transcriptomic features compared to the number of samples can cause instability of some machine learning models.

19. FUTURE SCOPE

Future enhancements may be made to include:

- The validation of the work was carried out using external datasets that were of GEO origin.
- Advanced Differential Expression Analysis with R/Bioconductor.
- survival analysis
- Gene Set Enrichment Analysis (GSEA)
- multi-omics integration
- deep learning-based classification
- drug target prediction
- The discovery of therapeutic biomarkers specifically for each subtype of the disease.

Additional research studies could also involve applications for personalized medicine and the development of prognostic biomarkers.

20. CONCLUSION

In this project the entire transcriptomic and machine learning workflow for molecular subtype analysis of breast cancer was successfully applied on the data platform Orange Data Mining and several external bioinformatics platforms.

Distinct molecular signatures for each of the subtypes were identified using pre-processing, PCA, clustering analysis, feature selection, machine learning classification, differential expression analysis, GO enrichment, KEGG pathway analysis, and STRING interaction analysis.

SVM appeared to be the best algorithm among the evaluated machine learning algorithms, and the effectiveness of the transcriptomic biomarkers in predicting subtypes was confirmed.

Integrated analyses of systems biology further identified a close relationship between HER2-positive breast cancer and the dysregulated pathways of the spindle assembly, chromosome segregation, and proliferative cell cycle pathways, as well as dysregulated signaling of the mitotic checkpoint.

Although this was my first transcriptomics project, it gave me some great hands-on experience with bioinformatics, finding biomarkers, machine learning, and analysis of cancer systems.